

Dynamic Commodity Timing Strategies

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ABSTRACT

Recent research documents that commodities are good diversifiers in traditional investment portfolios: overall portfolio risk is reduced while less than proportional return is sacrificed. These studies generally find a relatively high volatility in commodity returns, which implies a huge potential for tactical strategies. In this paper we investigate timing strategies with commodity futures using factors directly related to the stance of the business cycle, the monetary environment and the sentiment of the market. We use a dynamic model selection procedure in the spirit of the recursive modeling approach of Pesaran and Timmermann [1995]. However, instead of using in-sample model selection criteria, we build on the extensions of Bauer, Derwall and Molenaar [2004] by introducing an out-of-sample model training period to select optimal models. The best models from this training period are used to generate forecasts in a subsequent trading period. Our results show that the variation in commodity future returns is sufficiently predictable to be exploited by a realistic timing strategy.

For a long time, commodities were deemed inappropriate investments because of their perceived risky character. The disappointing performance and future prospects of traditional asset classes and the availability of data and commodity indices have rapidly changed this situation. Moreover, a number of studies recently confirmed that adding commodities to a balanced portfolio of more traditional assets reduces overall risk, despite substantial stand-alone risk. Commodities actually serve as diversifiers: overall portfolio risk is reduced, while none or less than proportional portfolio return is sacrificed (for examples, see Abanomey and Mathur [2001], Ankrim and Hensel [1993], Anson [1999], Becker and Finnerty [1994], Georgiev [2001] and Kaplan and Lummer [1998]). Edwards and Caglayan [2001] show that commodity funds have higher returns during bearish stock markets, along with a lower correlation. Related to this, Chow, Jacquier, Kritzman and Lowry [1999] provide evidence that commodities perform well when the general financial market climate is negative. Furthermore, commodities appear to serve as a possible hedge against inflation, see Bodie [1983], Froot [1995] and Gorton and Rouwenhorst [2004], which makes them even more attractive for entities with fixed liabilities in real terms, like for instance pension funds. Finally, Nijman and Swinkels [2003] show that commodity investments are beneficial to pension funds within a mean-variance framework.

Based on this evidence institutional investors are increasingly integrating commodities in their strategic asset allocation, predominantly in a passive fashion. Although the literature on the strategic benefits of investing in commodities is growing, papers on tactical asset allocation with commodities are quite difficult to find. Notable exceptions are the work of Johnson and Jensen [2001] and Jensen, Johnson and Mercer [2002] in which the allocation to commodities is conditioned on the monetary environment. Furthermore Nijman and Swinkels [2003] recently examined a tactical switching strategy between commodities and stocks. Most of these studies use a small set of predetermined explanatory variables to base their tactical decisions on. In contrast, we will use a dynamic, multi-factor approach to forecast monthly commodity returns using a broad universe of macro-economic and (market) sentiment indicators. These forecasts are subsequently exploited in a realistic market timing strategy.

We first present our base set of candidate predictor variables. Subsequently, we introduce a dynamic modeling approach proposed by Pesaran and Timmermann [1995]. Although we apply their methodology in a similar way, we include the methodological adjustments recently put forward by Bauer, Derwall and Molenaar [2004] in an equity style timing context. In the empirical part we provide results of a commodity market timing strategy and a variety of robustness checks and sensitivity analyses. It appears that this strategy is capable of generating information ratios well above a benchmark strategy of simply buying and holding commodity futures. In the last section we illustrate how the timing strategies perform when the model selection routine is conditioned on the portfolio manager's ex ante macroeconomic beliefs.

COMMODITIES AND THE MACRO-ECONOMY

The notion that commodity futures returns are related to the macro-economy is supported by Strongin and Petsch [1995, 1996] and more recently by Gorton and Rouwenhorst [2004]. Gorton and Rouwenhorst study the properties of commodity futures as an asset class. They show the behavior over the business cycle and the positive relation of commodity returns with inflation. Strongin and Petsch also find that commodities, in sharp contrast to traditional assets, are more directly linked to current economic conditions. As the level of economic activity increases, expected returns for commodities tend to rise. According to Strongin and Petsch this strong link with the macro-economy - relative to other asset classes - provides a good opportunity for a timing strategy. Building on this notion, we construct a broad set of explanatory variables with a strong link to the business cycle, the monetary environment and financial markets' sentiment.

In this study we aim at forecasting the direction of monthly returns of the Goldman Sachs Commodity Index (henceforth: GSCI). The GSCI is a passive, tradable buy-and-hold index of 25 commodities (ultimo December 2003). Futures on this index are screened on their liquidity and relevance in terms of their weight in the world production. Exhibits IA

and 1B show the cumulative return and summary statistics of the GSCI in the past three decades.

<< Please insert exhibits 1A and 1B around here >>

There are no obvious patterns in the series, but it appears that the variability is particularly high during the oil-crises of the seventies and the Gulf-war, and in the most recent turbulent period. The shaded areas in exhibit 1A are NBER indicated recession periods. Total returns during recession periods are slightly lower than returns in booming periods. Our unpublished results, as well as Gorton and Rouwenhorst [2004] show that commodity returns are in general above their average during late expansion and early recession periods of the business cycle, exactly when stocks and bond returns are below their overall average. The mean annualized total return of the GSCI during the full sample period is 12.90%. The standard deviation is relatively high: 18.42%. Monthly minimum and maximum returns of -15.64% and 25.77% show that commodities may be considered risky in a stand-alone context. On the other hand, the volatility in the GSCI series implies that there is a huge potential for timing strategies.

Our base set of explanatory variables consists of three classes linked to the existing academic timing literature: (1) business cycle indicators, (2) monetary environment indicators and (3) indicators on the (market) sentiment. These variables have been used predominantly in studies investigating the link between the (macro-) economy and traditional asset classes, or in timing studies like for instance Pesaran and Timmermann [1995]. To the best of our knowledge, with the exception of the monetary environment dummy of Jensen, Johnson and Mercer [2002], none of these indicators have been used in a commodities timing framework. Given the nature of our candidate variables and availability issues, we restrict our attention to U.S. data.

With respect to the class of business cycle indicators, Chen [1991] shows that the dividend yield and the default spread are (inversely) related to current business cycle conditions. In our models we include the (annualized) *dividend yield on the S&P 500* and

the (annualized) *yield spread between long-term Moody's rated BAA- and AAA-bonds*. Moreover, Chen indicates that the one-month Treasury bill and the term spread are related to more distant business cycle conditions. Our term spread variable is constructed as the *difference in yields between a constant maturity 10 Year T-bond and a 3-month constant maturity T-bill*. Finally, Chen finds a positive link between the business cycle and annual production growth and GNP (and consumption). We therefore include the *change in year-over-year industrial production*.

Bodie [1983], Froot [1995], Strongin and Petsch [1996], Jensen, Johnson and Mercer [2002] as well as Gorton and Rouwenhorst [2004] explicitly document the inflation hedging properties of commodities. To capture this insight we include the *year-over-year rate of inflation* in our database. Jensen, Johnson and Mercer [2002] show that the monetary environment is helpful in discriminating between good and bad commodity performance. We follow their classification to characterize the monetary situation and construct a *discount rate dummy*. This variable has value zero (one) when the monetary situation is expansive (restrictive). If the last change in the Federal Reserve discount rate was a decrease, the regime is indicated as expansive (: value 0). Similarly, if the last change was an increase, the regime is classified as restrictive (: value 1). We additionally include *monetary aggregate M2* in our set of regressors.

The sentiment on the stock market is usually seen as a (be it noisy) predictor of future economic developments. For this reason we add the *total returns of the S&P 500* to the database. The *one-month lagged GSCI return*, the *average GSCI return over the last 12 months* and *previous 36-month GSCI standard deviation* are selected in order to account for possible momentum in commodities markets. In order to capture variables linked to the sentiment of the economy in general, we include *year-over-year changes in consumer- and business confidence*. Finally, with the U.S. being the major commodity consumer and as most commodities are listed in U.S. dollars, we select the *trade weighted U.S. dollar*.

We downloaded all explanatory variables from Datastream and implemented appropriate lags to take publication lags in the macro series into account.

A DYNAMIC MODELING APPROACH

The ability to time asset classes is the backbone of many supposedly feasible timing strategies. Unfortunately, despite overwhelming evidence from the academic literature, the benefits of predictability are hardly observed in practice. As pointed out by Cooper and Gulen [2002], the apparent predictability gap might be due to substantial biases in many reported findings obtained from a setting that benefits too much from *ex post* knowledge. A classic example is the estimation of a single predictive model based on the entire sample period, which is not obtainable by investors in “real time”. Although many papers validate the predictive ability by applying an out-of-sample framework, many other parameters, including the choice of predictive model, are usually determined with the benefit of hindsight. In order to obtain truly practical results with such a procedure, the assumption of a time-invariant joint significance of the determinants needs to hold. This is very doubtful. Provided the empirical results in the back-testing process rely substantially on these parameters, the economic significance will be exaggerated. To mitigate the impact of “hindsight” bias, we simulate our trading strategies by means of a dynamic modeling approach in which we explicitly account for the continuous uncertainty that real-time investors face concerning the choice of the optimal set of predictive variables.

Our procedure is largely an extension of the work of Pesaran and Timmermann [1995], who introduced the approach at the stock market return predictability level for the United States. Using our base set of forecasting variables we first define a universe of parsimonious models based on in-sample estimation. Following this, we allow for the selection of a “best” model according to a predefined selection criterion. However, whereas most studies use in-sample model selection criteria, we increase the likelihood of a successful forecast by introducing an *out-of-sample training* period to test and select

our models. We start the implementation of the timing strategy in a second-stage out-of-sample period, from hereon referred to as the *trading* period. All events re-occur on a monthly basis via a rolling window framework. The choice of a selection period that postdates the model estimation sample relates to the evidence of Bossaerts and Hillion [1999] who failed to find sufficient out-of-sample predictability when using conventional in-sample selection criteria.ⁱ

In essence, the model selection procedure is aligned with the ultimate objective of any forecasting model in practice: a high *realized* information ratio (IR). In the context of our commodities timing strategy we use a 60-month in-sample estimation window and a 24-month training period. In order to obtain parsimonious model specifications, we restrict the set of explanatory variables in the models to be between 0 and 5 (excluding a constant). This eventually leaves us with 4,944 out of 32,768 ($= 2^{15}$) possible models. During the in-sample period, we estimate parameters for these models using OLS. Following this, each model generates monthly signals during a 24-month training periodⁱⁱ. In the case of a positive signal for commodities, regardless of the strength, we buy futures on the GSCI-index and in the case of a negative signal we sell these futures. At the end of the training period we rank all models on realized information ratios having taken into account transaction costs. The strategy with the highest realized information ratio is used to forecast the sign of next month's GSCI index return. Finally, in the out-of-sample trading period we buy or sell futures on the GSCI index dependent on the signal.

This procedure is repeated every month (see exhibit 2) and generates a ranking of preferred models for every time-period in the sample and subsequent out-of-sample timing decisions. Models are thus dynamically re-estimated and re-selected every month, which is in accordance with investors continuously searching for the best model specification given their data at that point in time. As a yardstick to measure the success of our timing strategy in the trading period we compare the returns with a buy-and-hold commodity strategy.

<< Please, insert exhibit 2 around here >>

EMPIRICAL RESULTS

Exhibit 3 shows the results of both the buy-and-hold (BH) portfolio and the timing strategy. The GSCI index was introduced in July 1992. Our out-of-sample trading period therefore starts in August 1992 and ends in December 2003. Because the index was backfilled to 1970, we restrict our attention to a sample period in which this timing strategy could be pursued in real time. The annualized mean excess return of the BH strategy is 2.94% versus 11.80% for the timing strategy assuming no transaction costs. Annualized standard deviations are comparable (18.30% versus 18.00%). This leads to an IR of 0.16 for the BH strategy versus 0.66 for the timing strategy. We find that the IR of the strategy is significantly different from zero, based on the approach of Lo [2002] in calculating the standard error of the IR. The hit-ratio defined as the percentage of correctly predicted signals is 60%. According to the Henriksson-Merton [1981] non-parametric market-timing test, the active strategy possesses significant timing skill at the 5%-level of significance. Obviously, the buy-and-hold strategy is long 100% of the time, whereas the active strategy is long in roughly 61% of the months and short in 39% of the months.

<< Please, insert exhibit 3 around here >>

Exhibit 3 additionally provides information on the impact of transaction costs. Suppose, for example, that we have 2 models: X and Y. If model X has a higher training period IR than model Y before transaction costs, but a lower one after transaction costs, we select model Y. Using this procedure, we explicitly punish models that trade often and thus incur higher transaction costs. Since we already take into account transaction costs in the training period, different models may be selected when assuming different transaction cost scenarios. We calculate the performance under 3 transaction cost scenarios: 10, 25 and 50 basis points single trip. It should be noted that these transaction costs can be seen as *incremental*. Because futures have a finite life, a buy-and-hold strategy with futures

also incurs transaction costs. We do not account for transaction costs for the buy-and-hold strategy and consider the transaction costs for the strategy as additional to what a buy-and-hold investor would incur. Exhibit 3 shows that the timing strategy suffers from higher transaction costs, as one may expect. The drop in IR is however not dramatic and even in the case of high transaction costs (50 basis points) the active strategy remains attractive.

The upper panel in exhibit 4 shows the cumulative returns of the BH and the timing strategy. The lower panel plots corresponding positions (“long” or “short” in commodities) over time. Until the end of 1997, the active strategy performs marginally better than the buy-and-hold strategy. The severe commodity market downturn that kicks in at the beginning of 1998 is however well anticipated by the strategy. When the market starts to recover in 1999, short positions are timely transformed into long positions. Although the strategy does not realize substantial outperformance during the first part of the sample, it is reassuring that it takes correct positions during major cyclical market moves. During the last 5 years of the sample, the strategy performs quite well. The overall hit-ratio of 60% illustrates that the performance of the strategy is not the result of just of a few “lucky shots”.

<< Please, insert exhibit 4 around here >>

A natural question that arises is which variables are predominantly selected over time. Exhibit 5 plots the factor inclusion over time and exhibit 6 provides additional sub-period information. Both exhibits show that variable inclusion is not stable over time, justifying the dynamic approach we follow. Over the whole sample period a few factors are included in the timing models frequently: *S&P 500 return*, *Business Confidence*, *Industrial Production*, *M2* and the *U.S.-Dollar*.ⁱⁱⁱ From exhibit 4 we learned that the timing strategy successfully anticipated both the market downturn in 1998 and the subsequent upswing from 1999 to the end of 2000. Looking at the information in exhibit 6 we can identify that the dynamic modeling approach in the sub-period (1998:01 – 2000:12) mainly selected business cycle and sentiment variables and virtually none of the

monetary variables. For instance, in this period the relative weight of the *U.S. Consumer Confidence* indicator was more important and the *U.S.-Dollar* was included much less relative to the full sample. In the last three years of the sample monetary indicators seemed to be more relevant again, while reducing the weight of the *U.S. T-bill*. These swings in the selection of explanatory variables would not have been possible in static timing models.

<< Please, insert exhibits 5 and 6 around here >>

ROBUSTNESS AND SENSITIVITY ANALYSIS

The previous section showed that our dynamic modeling approach is capable of outperforming a strategy that simply buys and holds the GSCI. In this section we investigate how sensitive the timing strategy is to changes in the model settings. We first re-run the model selection procedure where we calculate next month's forecast as a weighted average of the top n models instead of solely using the forecast of the highest ranked model. Secondly, we examine whether the model performance is largely driven by the heavy representation of energy in the GSCI. It may be that the model is especially capable of forecasting “energy” or “agriculture” instead of a broad basket of commodities. We therefore re-run the model on all GSCI sub-indices.

So far the timing strategy has generated signals based on the single best performing model in the training period. If we do not want to be dependent on one (outlier) model, we alternatively could select the top n models from the training period. These n models are averaged to provide us with a forecast for next month (i.e. the trading period). To take the relative strength of the signals into account, we calculate next month's forecast as a weighted average of the top n models. Let n^+ denote the number of models with a positive and n^- with a negative forecast for next month's return. We calculate the aggregate position for next month as: $(n^+ - n^-) / n$. For example, suppose we average over

the 25 top models from the training period. From these 25 models, 15 give a buy signal and 10 a sell signal. In this case, we take a 20% long position (i.e. $(15 - 10) / 25$).

Exhibit 7 shows the information ratio of the timing strategy as a function of the number of models averaged over. It appears that the performance of the strategy fluctuates quite a bit for the first couple of models (IRs between 0.6 and sometimes even 0.9). The general trend is that the information ratio of the strategy ultimately declines as a function of the number of models averaged over. This is what we expected: models with a lower ranking during the training period perform less than models with a higher ranking. The relatively modest decline in performance stems from the fact that we take a weighted average of all models. So, whereas averaging over 2 instead of 1 model may result in a large swing in the forecast, averaging over 2000 instead of 1999 models does much less so. The general conclusion from this analysis is that an information ratio between 0.65 and 0.70 can be achieved even after averaging over the first 1 to 1000 top performing models (in the default case of a maximum of 5 variables)^{iv}.

<< Please, insert exhibit 7 around here >>

The GSCI consists primarily of energy related commodities: 67% of the index weight (ultimo 2003). The performance of the model may be caused by its ability to forecast energy futures returns correctly rather than across a broad spectrum of different commodities. To investigate this we use the same set of variables to forecast the different GSCI sub-indices: *Agriculture*, *Energy*, *Industrial Metals*, *Livestock*, and *Precious Metals*. Exhibit 8 shows that the timing strategy adds value in all cases with the exception of *Livestock*. Statistically significant timing possibilities are limited to the *Energy* sub-index and, to a lesser extent, the *Industrial Metals* sub-index.

<< Please, insert exhibit 8 around here >>

CONDITIONING ON ECONOMIC INTUITION

Besides robustness issues, we want to take into account a particular criticism of the dynamic approach. The issue many portfolio managers might have with this approach is the lack of economic ratio supporting the model. Although the selected variables are possibly related to the business cycle, the dynamic modeling approach does not incorporate economic theory. A portfolio manager may for example, based on previous empirical evidence or on his personal view, wish to restrict the sign of the business cycle variables in the model to be positive. Another example in this context is related to the default-spread, which is thought to have a negative relation with future economic performance. Model specifications with counterintuitive signs, although optimal in a statistical sense, should then not be taken into account. The basic thought behind this is that erroneous short-run dynamics are probably specific for the time period considered and may end as soon as they came. Having an incorrect and “unexplainable” model may then be the price if these non-regular relations suddenly disappear.

With regard to incorporating economic theory, we restrict the signs of variables with well-documented economic interpretations. We freely estimate all possible models, but take into account only those models that have coefficients in accordance with economic theory. Continuing with our example: model specifications with a positive coefficient on the default spread are disregarded^v. We restrict the signs for a set of variables for which the link with macroeconomic developments has been documented before. Following the results of Chen [1991], we restrict the signs on the *dividend yield*, the *default spread* and the *T-bill rate* to be negative and the signs on the *term spread* and *industrial production* to be positive. From the evidence in Bodie [1983] and Johnson, Jensen and Mercer [2002], betas on *core inflation* and the *discount rate dummy* should be positive. For the remaining variables the economic interpretations are less clear-cut and we leave those unrestricted.

<< Please, insert exhibit 9 around here >>

Exhibit 9 shows that imposing the restrictions found in the literature increases the performance slightly (IR goes up from 0.66 to 0.74). Further analysis of the factor inclusion over time (not reported) shows that the T-bill rate has an incorrect sign for all instances when it belongs to the top model. A (possibly unwanted) effect of imposing these restrictions is that this variable is therefore never included in the optimal model. As model performance is quite robust in the face of the imposed restrictions, the question whether or not to restrict the model therefore comes down to a basic trade-off: we could either “let the data freely speak its own language” with possibly only fitting short-term noise or imposing “firm economic beliefs” not necessarily found in the data. In our opinion, final judgment should be based on what the portfolio manager is most comfortable with.

CONCLUSION

In recent times, institutional investors have started to add commodities to their strategic asset mixes. Due to the low correlation with more traditional assets, overall risk is reduced while none (or less than proportional) return is sacrificed. We take a tactical asset allocation perspective. Using variables related to the business cycle, the monetary environment and market sentiment we build dynamic timing strategies. Instead of focusing on in-sample criteria, we use an out-of-sample training period to select the optimal model. The best performing model, in terms of realized information ratios during the training period, is employed to generate a forecast for the trading period. We show that the predictable variation in futures returns is sufficient to be exploited by a realistic timing strategy. Changing the number of optimal models averaged across and taking into account transaction costs does not alter this conclusion. Testing the model on the sub-index level showed that especially the *Energy* and *Industrial Metals* sub-indices are predictable. Finally we showed how portfolio managers can restrict the model to have “economically intuitive” coefficients. For this particular set of restrictions, model performance was even slightly better than for our base-case setting. Summarizing, it appears that investors can profit from tactical asset allocation with commodities in real-

time. The timing strategy delivers superior investment returns, both in an economical and a statistical sense.

ENDNOTES

The views expressed in this paper are from the authors and are not necessarily shared by their employer. We are very grateful to our colleagues at ABP Investments for stimulating discussions and comments on earlier versions of this article. We especially thank Jean Frijns for providing invaluable analytic and conceptual support as well as his ideas on conditioning on economic intuition. Comments by K. Geert Rouwenhorst, Yale University, Luis M. Viceira, Harvard University and Peter C. Schotman, Maastricht University greatly improved the quality of this paper.

EXHIBIT 1A
Cumulative Monthly Excess Returns of the Goldman Sachs Commodity Index:
1970:1 – 2003:12

Shaded areas are NBER indicated-recession periods. Annualized GSCI Total Return during recessions:
11.48%, during non-recession periods: 13.20%.

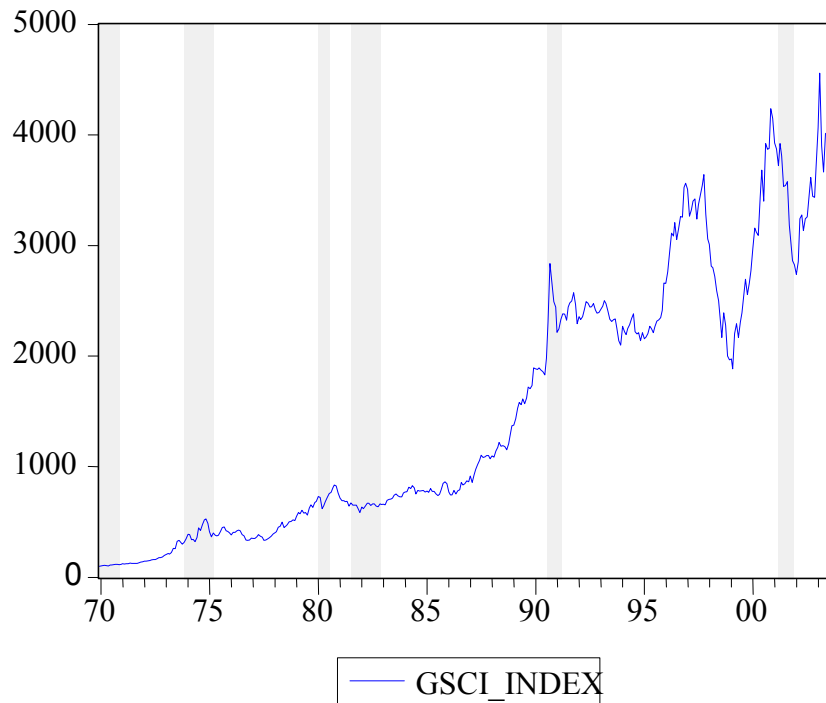


EXHIBIT 1B
GSCI Total Return Index characteristics: 1970:1 – 2003:12

	GSCITOT
Mean	12.90
Median	0.93
Maximum monthly	25.77
Minimum monthly	-15.64
Std. Dev.	18.42
Skewness	0.56
Kurtosis	5.63

EXHIBIT 2
The Dynamic Modeling Approach Graphically

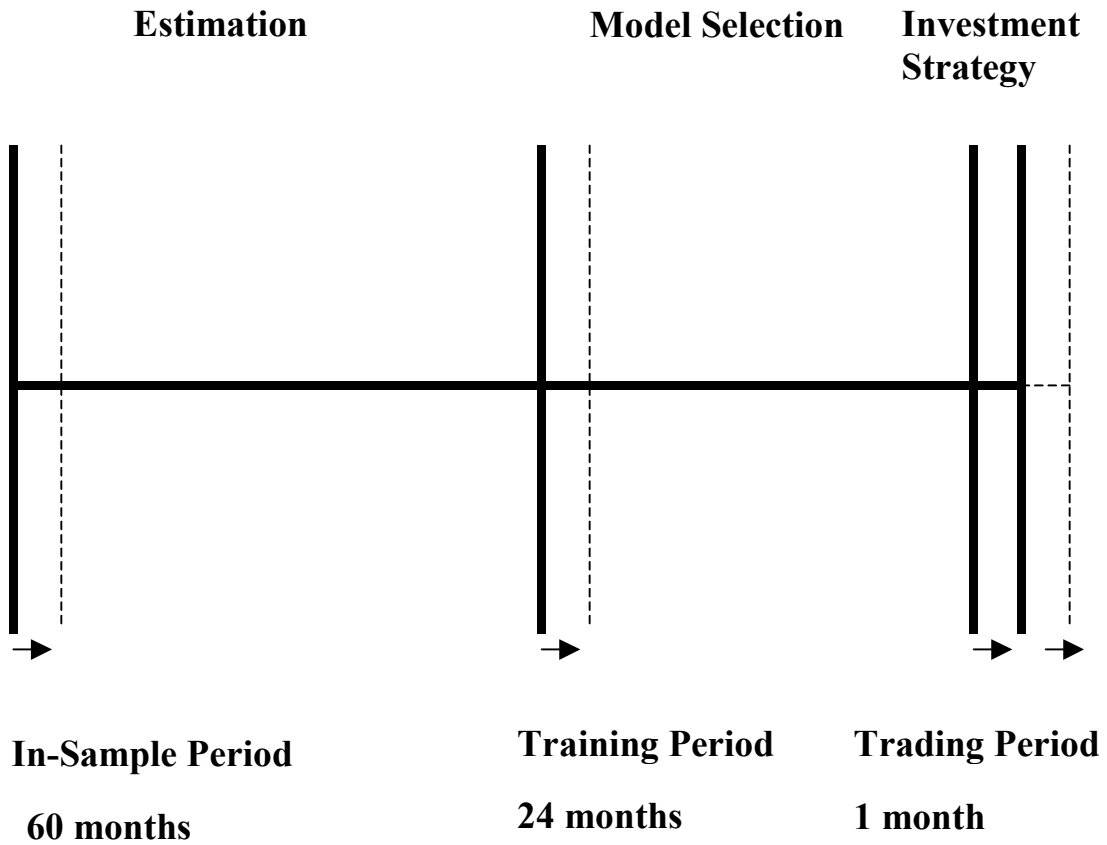


EXHIBIT 3

Summary Performance Statistics for Buy-and-Hold and Tactical Strategies under Various Transaction Cost Scenarios for 1992:8 – 2003:12

This exhibit shows the (annualized) performance statistics for the buy-and-hold strategy and the commodity timing strategy. Signals are from the optimal model in the training period. This training period consists of 24 months, whereas the estimation period is 60 months. The minimal number of variables is 0, the maximum 5. *, ** and *** indicate significantly different from zero at the 10-, 5 or 1%-level, respectively.

	BH	0 bp.	10 bp.	25 bp.	50 bp.
Mean Excess Return	2.94	11.80	10.88	9.97	8.65
Standard Deviation	18.30	18.00	18.02	18.04	18.09
Information Ratio	0.16	0.66 **	0.60 **	0.55 **	0.48 *
Median	0.08	1.11	0.97	0.97	0.87
Minimum	-14.49	-13.32	-13.32	-13.32	-13.50
Maximum	16.40	16.40	16.20	15.90	15.40
Hit Ratio		0.60 **	0.59 **	0.58 **	0.58 **
Months Long	100.00	61.31	62.04	61.31	61.31
Months Short	0.00	38.69	37.96	38.69	38.69

EXHIBIT 4

Cumulative Performance of the Buy-and-Hold Strategy and the Unrestricted Model

The upper panel of this exhibit shows the cumulative excess returns for the buy-and-hold strategy and the unrestricted commodity timing strategy. No transaction costs are taken into account. The lower panel provides the aggregate positions of the active strategy taken in the GSCI.

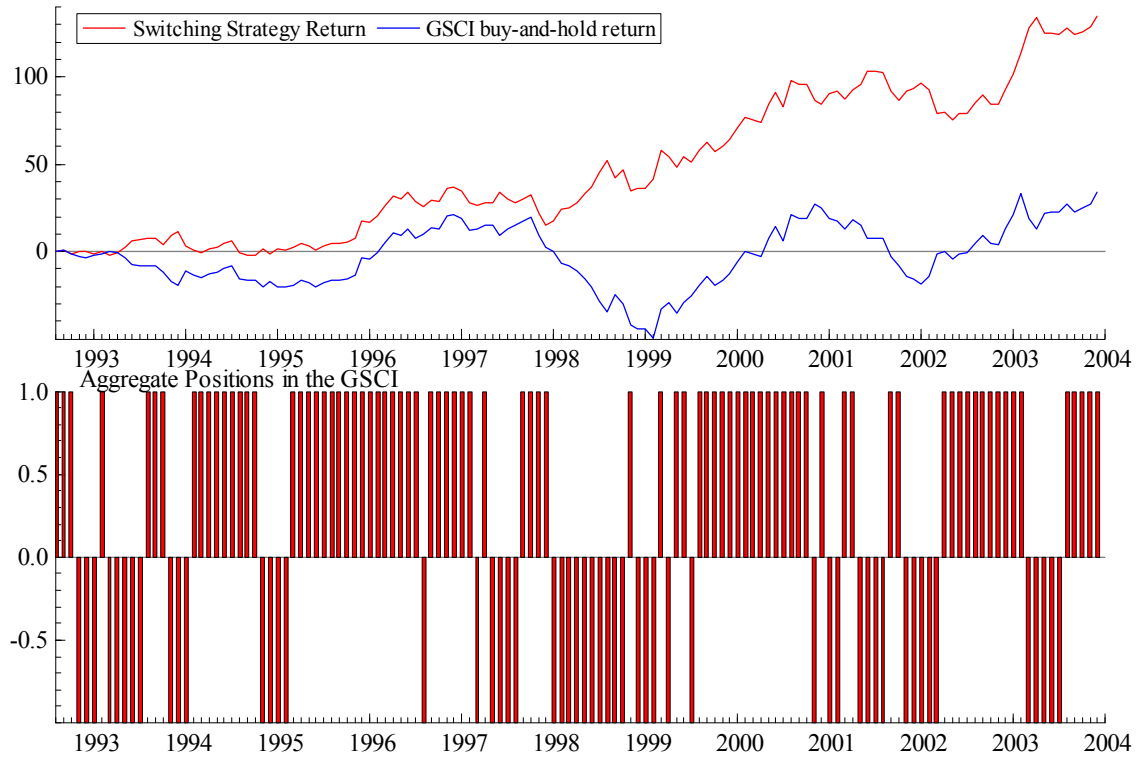


EXHIBIT 5

Factor Inclusion over Time

Below the inclusion in the optimal model of the 15 factors in every time period is displayed. Total inclusion in percentages is mentioned in parentheses.

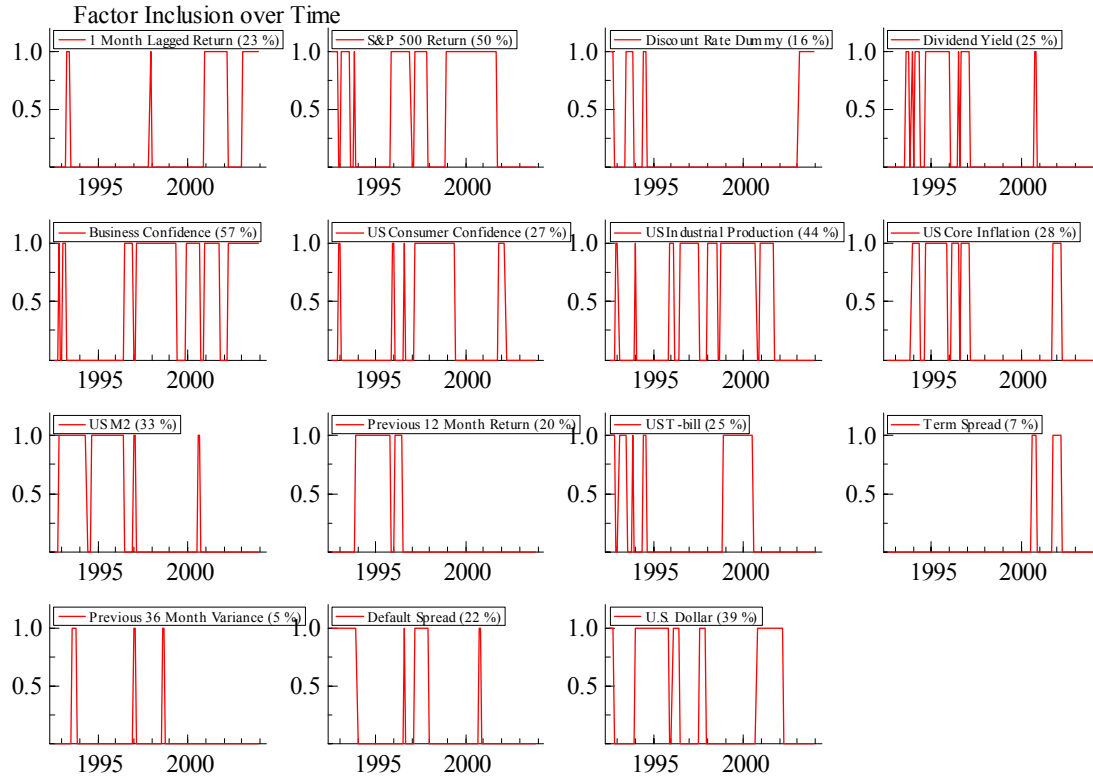


EXHIBIT 6

Factor Inclusion over Sub-periods

For every class of forecasting variables the inclusion in three sub-periods is shown as well as for the full sample

	Full-sample	1992:8 – 1997:12	1998:1 – 2000:12	2001:1 – 2003:12
Business Cycle Indicators				
Dividend Yield	25%	49%	6%	0%
U.S. Industrial Production	44%	29%	89%	25%
U.S. T-bill	25%	22%	56%	0%
Term Spread	7%	0%	11%	17%
Default Spread	22%	43%	6%	0%
1 Month Lagged Return	23%	6%	3%	72%
Monetary Indicators				
Discount Rate Dummy	16%	17%	0%	31%
U.S. Core Inflation	28%	49%	0%	17%
U.S. M2	33%	66%	6%	0%
Sentiment Indicators				
Business Confidence	57%	29%	78%	86%
S&P 500 Return	50%	52%	69%	25%
U.S. Consumer Confidence	27%	23%	47%	14%
Previous 12 Month Return	20%	43%	0%	0%
Previous 36 Month Variance	5%	8%	6%	0%
U.S Dollar	39%	54%	8%	42%

EXHIBIT 7

Information Ratio as a Function of the Number of Models Averaged Over

This graph shows the information ratio (IR) for the number of optimal models averaged over in the case of a maximum number of forecasting variables of 5.

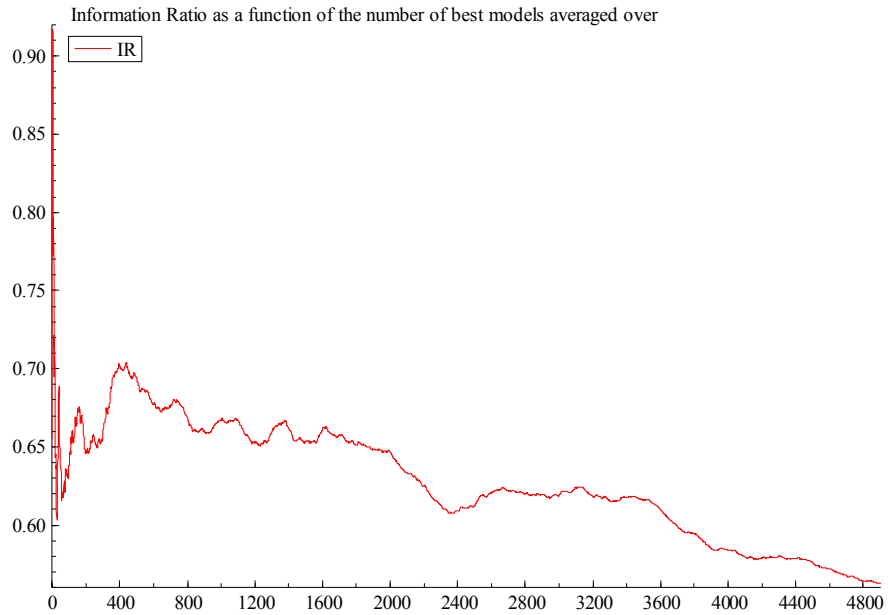


EXHIBIT 8

Summary Performance Statistics for Buy-and-Hold and Tactical Strategies for Sub-Indices

This exhibit shows the (annualized) performance statistics for the buy-and-hold strategy and the commodity timing strategies for the sub-indices. Signals are from the optimal model in the training period. This training period consists of 24 months, whereas the estimation period is 60 months. The minimal number of variables is 0, the maximum 5. *, ** and *** indicate significantly different from zero at the 10-, 5 or 1%-level, respectively.

<i>Weight in GSCI:</i>	Agriculture (17%)		Energy (67%)		Industrial metals (7%)	
	BH	Tact.	BH	Tact.	BH	Tact.
Mean Excess Return	-3.43	3.49	7.7	16.11	-1.08	6.46
Standard Deviation	13.97	13.87	30.27	30.00	15.98	15.88
Information Ratio	-0.25	0.25	0.25	0.54 **	-0.07	0.41 *
Median	-0.51	0.32	-0.13	0.75	-0.07	0.43
Minimum	-10.03	-10.03	-22.44	-34.13	-13.29	-13.29
Maximum	10.76	10.76	34.13	23.22	13.21	11.70
Hit Ratio		0.50		0.55 *		0.56 *
Months Long	100.00	45.26	100.00	56.20	100.00	41.61
Months Short	0.00	54.74	0.00	43.80	0.00	58.39

	Livestock (7%)		Precious Metals (2%)	
	BH	Tact.	BH	Tact.
Mean Excess Return	-3.53	-5.09	-0.35	2.01
Standard Deviation	14.14	14.1	12.62	12.61
Information Ratio	-0.25	-0.36	-0.03	0.16
Median	-0.17	-0.49	-0.49	-0.01
Minimum	-15.83	-15.83	-8.83	-8.92
Maximum	10.29	10.81	15.13	15.13
Hit Ratio		0.45		0.50
Months Long	100.00	56.20	100.00	45.99
Months Short	0.00	43.80	0.00	54.01

EXHIBIT 9
Summary Performance Statistics for Tactical Strategies on the GSCI, Imposing
Economic Intuition: 1992:8 – 2003:12

This exhibit shows the (annualized) performance statistics for the buy-and-hold strategy and the commodity timing strategy imposing economic intuition. The active strategy does not take into account transaction costs. Signals are from the optimal model in the training period. This training period consists of 24 months, whereas the estimation period is 60 months. The minimal number of variables is 0, the maximum 5. *, ** and *** indicate significantly different from zero at the 10-, 5 or 1%-level, respectively.

<i>no transaction costs</i>	BH	Tactical Strategy	
Mean Excess Return	2.94	13.24	
Standard Deviation	18.30	17.91	
Information Ratio	0.16	0.74	***
Median	0.08	1.31	
Minimum	-14.49	-16.40	
Maximum	16.40	14.80	
Hit Ratio		0.58	**
Months Long	100.00	56.20	
Months Short	0.00	43.80	

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ⁱ Conventional statistical criteria include the adjusted R^2 , the Aikake information criterion and the Schwarz criterion.

ⁱⁱ Choosing the appropriate length of the training period is somewhat arbitrary. On the one hand we need a period long enough to be able to evaluate the performance of the timing strategy, but on the other hand we cannot make this period too long as the estimated models could become less relevant as time proceeds.

ⁱⁱⁱ Note that the expected % of inclusion in the case of a maximum of 5 explanatory variables is approximately 30%.

^{iv} We additionally analyzed the IR as a function of the number of models averaged over for a maximum of 4 and 6 explanatory variables. Results are qualitatively the same and available upon request.

^v The way we impose this restriction is very strict. This can be done subtler within a Bayesian framework.